

UAV-Relaying Strategies for Free-space Quantum Key Distribution Systems

— Research Progress Report —
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Sota Fujii
s1310106 4-year undergraduate student
Computer Communications Laboratory,
The University of Aizu, Japan



会津大学

Outline

I. Introduction & Previous Research

- Motivation
- QKD Protocols
- Summary of results from previous work

II. New Research Phase (Research Update)

- Introduction of MDI-QKD
- Multi-Protocol Collaborative (MPC) QKD Networking
- Future research plan

Motivation: UAV-based QKD

Necessity of Quantum Key Distribution (QKD)

- Risk to Current Security: Conventional encryption (like RSA) could be broken by future quantum computers.
- QKD provides "Information-Theoretic Security" based on the quantum mechanics, which cannot be broken by any computer.

Necessity of UAV Integration

- No Cables Needed: In disasters or emergencies, fiber-optic cables are often broken or unavailable.
- Mobility & Flexibility: UAVs can fly to any location and act as mobile relay nodes to build a secure network.



Goal of this Research

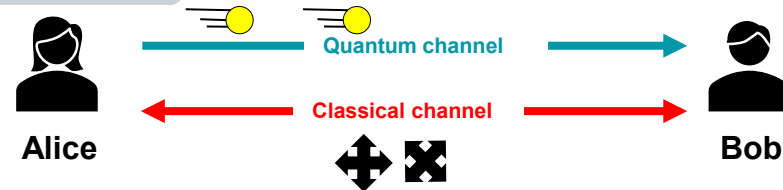
Secure & Flexible Network: Combining UAV mobility with QKD security to protect data anywhere, anytime.

Quantum Key Distribution Protocol

BB84 Protocol:

1. Alice sends single photons to Bob via a quantum channel.
2. They share measurement bases via a classical channel.
3. They keep the bits only when their bases match to form a secret key.

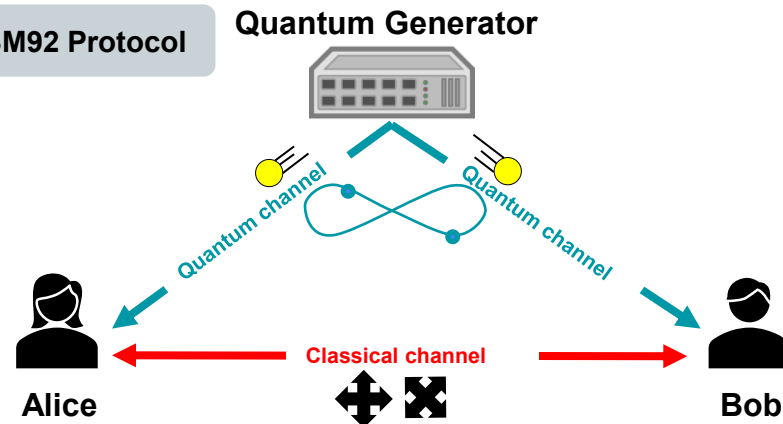
BB84 Protocol



BBM92 Protocol:

1. A source distributes entangled photon pairs to Alice and Bob via quantum channels.
2. They share their bases via a classical channel.
3. They keep the bits only when their bases match to form a secret key.

BBM92 Protocol



Previous Work

Evaluated 4 Relay Strategies:

- 1-UAV BB84
- 1-UAV BBM92
- 2-UAV BB84
- 2-UAV BB84 & BBM92

Key Results:

- **Distance-dependent Performance:** 1-UAV BBM92 is highest Kye rate for shorter distances (1–20 km).
- **Superiority of Hybrid Model:** The 2-UAV BB84 & BBM92 strategy achieved the highest key rate for long-distance QKD (beyond 20 km).

Conclusion:

These results helped me understand which strategy is better for each distance.

UAV-Relaying Strategies for Free-Space Quantum Key

Distribution Systems

Fujii Sota, Cuong T. Nguyen, and Hoang D. Le

Computer Communications Laboratory, The University of Aizu, Japan



I. Introduction

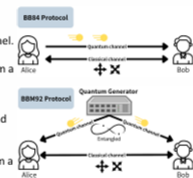
Motivation: Conventional encryption is threatened by quantum computers. A potential solution is Quantum Key Distribution (QKD). A secure communication method that uses quantum mechanics to guarantee secret keys.

BB84 Protocol:

1. Alice sends single photons to Bob via a quantum channel.
2. They share measurement bases via a classical channel.
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BBM92 Protocol:

1. A source distributes entangled photon pairs to Alice and Bob via quantum channels.
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3. They keep the bits only when their bases match to form a secret key.



The Problem: Terrestrial Free-Space Optics (FSO) links are often blocked by urban obstacles, and loss increases significantly over long distances. A potential solution is using Unmanned Aerial Vehicles (UAV) to extend the distance.



Objective: To address the issue, we compare different UAV-based relaying strategies based on BB84 and BBM92.

II. System Model

(a) 1-UAV BB84

In 1-UAV BB84, Alice and the UAV generate K1 (uplink), while the UAV and Bob generate K2 (downlink). The UAV then sends $M = K1 \oplus K2$ to Bob, who decrypts it to obtain the shared Secret Key K1.

(b) 1-UAV BBM92

In 1-UAV BBM92, the UAV distributes entangled photons to Alice and Bob, allowing them to directly generate a shared secret key without any encryption steps.

(c) 2-UAV BB84

Uses the BB84 protocol for all transmission links.

Key Sharing Procedure:

1. Generate keys K1, K2, K3 for each link using BB84.
2. UAV1: Send $M_a = K1 \oplus K2$ to UAV2.
3. UAV2: Decrypt M_a , then send $M_b = K1 \oplus K3$ to Bob.
4. Bob: Decrypt M_b with K3 to obtain the final key K1.

(d) 2-UAV BB84 & BBM92

Combines entanglement-based BBM92 (Alice-UAV2 via UAV1) with BB84 (UAV2-Bob).

Key Sharing Procedure:

1. UAV1: Distribute entangled photons to Alice and UAV2 to generate K1 (BBM92).
2. UAV2 & Bob: Generate K2 using BB84.
3. UAV2: Send $M = K1 \oplus K2$ to Bob.
4. Bob: Decrypt M to obtain the shared key K1.

Calculation of Secret Key Rate (SKR)

The SKR is defined as the average number of key materials that can be generated per second.

$$SKR_{total} = \min\{SKR_{link1}, SKR_{link2}, \dots, SKR_{linkn}\}$$

III. Results & Discussions

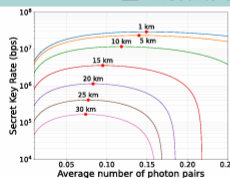


Fig 1: Optimal SKR of BBM92 at different communication distances.

The optimal emitted rate depends on the distance. We can select the optimal emitted rate to maximize the SKR at each distance.

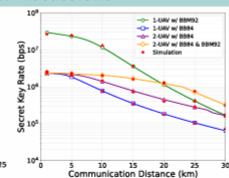


Fig 2: SKR of different strategies versus communication distances.

The theoretical values (solid lines) match the simulation results (red dots).

IV. Conclusion

- This research compares different UAV-relaying strategies to bypass obstacles in Free-Space QKD systems.
- We optimized the emitted rate of BBM92 protocols to maximize the SKR for each communication distances.
- 1-UAV BBM92 is best for 1–20 km, while 2-UAV BB84 & BBM92 is superior for distances beyond 20 km.
- These results provide a practical guideline for choosing the best UAV strategy, long-distance QKD systems.

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Research Update

Target Paper

“A Practical Multiprotocol Collaborative Quantum Key Distribution Networking Scheme for Consumer Electronics Devices”

Expanding Research Scope:

- Previous Focus: Performance comparison of BB84 and BBM92.
- New Focus: Introducing a **Multi-Protocol Collaborative (MPC)** framework.
 - Integrating **MDI-QKD (Measurement-Device-Independent QKD)** with existing protocols.
 - Goal: Balancing high communication capacity (via BB84) and untrusted relay security (via MDI).

A Practical Multiprotocol Collaborative Quantum Key Distribution Networking Scheme for Consumer Electronics Devices

Jia-Meng Yao[✉], Qiong Li[✉], Member, IEEE, Hao-Kun Mao[✉],
and Ahmed A. Abd El-Latif[✉], Senior Member, IEEE

Abstract—The development of emerging consumer technologies such as the Internet of Things (IoT), artificial intelligence (AI), and cloud computing brings convenience but also raises critical concerns about securing communication data. Quantum computing threatens the security of classical public key cryptography, making post-quantum data protection essential. Quantum Key Distribution (QKD) networks offer a solution, but the dependence on trusted relays raises security concerns. The existing QKD networks mainly consist of two types: measurement-device-dependent protocol based QKD networks which have a high dependence on trusted relays, and measurement-device-independent protocol based QKD networks which have a weak dependence on trusted relays but communication capability is limited. This paper proposes a Multi-Protocol Collaborative (MPC) networking cell combining both types of QKD protocol, which can reduce the dependence on trusted relays while maintaining a high communication capacity, and consequently has an enhanced practicality. An optimal topology method is also designed to further enhance the performance of the networks. Results show that communication capacity is improved 37 times compared to measurement-device-independent protocol based QKD networks. In addition, 23% reduction in the dependence on trusted relays compared to measurement-device-dependent protocol based QKD networks. By overcoming the trade-off between trust and capacity, this work significantly advances the practicalization of QKD networks in consumer electronics.

Index Terms—Quantum key distribution (QKD), QKD networks, multi-protocol collaboration, mathematical model, topology optimization.

I. INTRODUCTION

THE EVOLUTION of the Internet of Things (IoT), artificial intelligence (AI), edge computing, cloud computing, and other emerging consumer technologies has propelled consumer electronics into its next generation [1]. The next-generation consumer electronics exhibit higher connectivity between devices than traditional consumer electronics, leading to an exponential increase in cyber threats [2]. Moreover, next-generation consumer electronics are increasingly utilized in fields such as finance and healthcare, where data privacy and confidentiality are of utmost importance. It will lead to severe threat to the user's privacy if such data are accessed by malicious attackers [3], [4], [5]. Therefore, securing the confidentiality of sensitive data during communication is of great importance in next-generation consumer electronics [6].

A. Research of Communication Security in Consumer Electronics and QKD Solutions

To address the issue of communication security in consumer electronics, extensive researches have been conducted. For example, Kumar et al. explored blockchain-based authentication schemes for securing connected consumer electronics against wireless attacks [5]. Macedo et al. [7] and Kumar et al. [8] proposed device assessment frameworks for consumer IoT devices based on blockchain. Wortman et al. introduced an authentication scheme for consumer electronic devices utilizing various physically unclonable functions (PUFs) as a potential seed for a pseudorandom number generators (PRNGs) element [9]. Additionally, Golec et al. and Miao et al. developed authentication protocols for edge devices in IoT based on biometrics [10] and three-factor authentication [11], respectively. Furthermore, other scholars have investigated security schemes based on deep learning approaches for digital forensics [12], intrusion detection [13], and access control [14] in consumer electronics.

As shown from the reviewed literatures, the majority of existing researches on communication security in consumer electronics primarily focused on authentication issues, and data confidentiality protection still relies on public key encryption algorithms due to the difficulties of symmetric key distribution [15]. However, the development of quantum computing technology has seriously threatened the security of classical public key cryptography which is

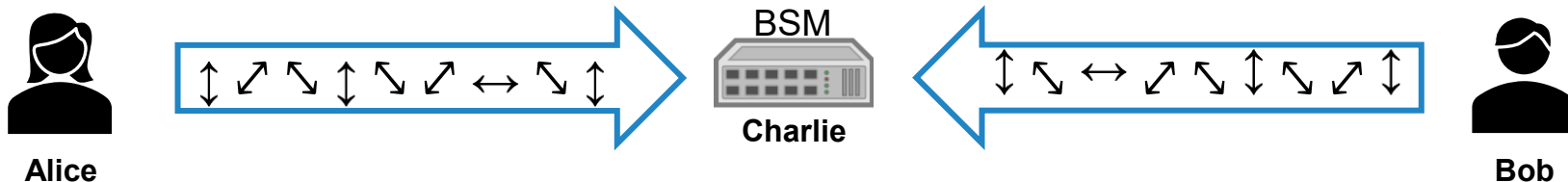
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Jia-Meng Yao, Qiong Li, and Hao-Kun Mao are with the Department of School of Computer Science and Technology, Harbin Institute of Technology, Harbin 150001, China (e-mail: 208903022@stu.hit.edu.cn; qiongli@hit.edu.cn; hkmiao@hit.edu.cn).

Ahmed A. Abd El-Latif is with the EHAS Data Science Lab, College of Computer and Information Sciences, and Center of Excellence in Quantum and Intelligent Computing, Prince Sultan University, Riyadh 11586, Saudi Arabia, and also with the Department of Mathematics and Computer Science, Faculty of Science, Menoufia University, Shibin El Khayma 6132501, Egypt (e-mail: abdelatif@psu.edu.sa).

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MDI-QKD(Measurement-Device-Independent QKD)

A relay-assisted QKD protocol where Alice and Bob establish a secure key through an untrusted node using Bell State Measurement (BSM).



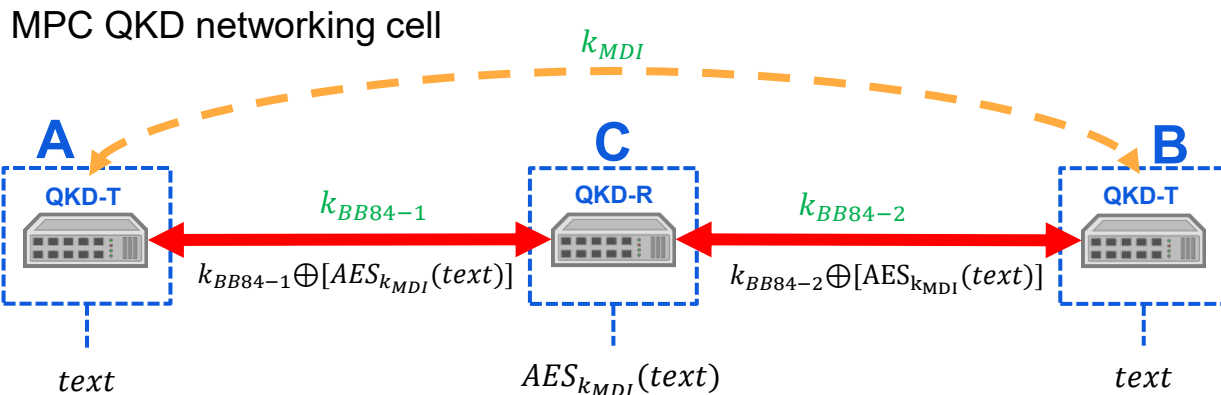
Alice's bit	0	1	1	0	0	1	1
Alice's Base	x	+	x	+	+	x	x
Qubits	$\nearrow$	$\updownarrow$	$\nwarrow$	$\leftrightarrow$	$\leftrightarrow$	$\nwarrow$	$\nwarrow$
Basis match	No	Yes	No	Yes	No	Yes	Yes
Bell State measurement match	No	Yes	No	No Bit Flip ↓	Yes	Yes	No Bit Flip ↓
Sifted key	-	1	-	1	-	1	0

Bob's bit	0	1	1	1	0	1	0
Bob's Base	+	+	+	+	x	x	x
Qubits	$\leftrightarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\leftrightarrow$	$\nwarrow$	$\nearrow$
Basis match	No	Yes	No	Yes	No	Yes	Yes
Bell State measurement match	No	Yes	No	No	Yes	Yes	No
Sifted key	-	1	-	1	-	1	0

Multi-Protocol Collaborative (MPC) QKD Networking cell

Concept:

A hybrid encryption scheme that collaboratively utilizes two types of QKD protocols: BB84 and MDI-QKD.



Key Exchange and Encryption Workflow

1. MDI-key(k_{MDI}): Generated between end-nodes(A&B) using an untrusted relay. It is used for AES encryption to protect data at the relay node.
2. BB84-key(k_{BB84}): Generated between adjacent nodes (A-C and C-B). It is used for OTP encryption to secure the transmission channels.
3. Hybrid Ciphertext: The final data is transmitted as $k_{BB84} \oplus [AES_{k_{MDI}}(text)]$.

Future Outlook & Plan

Integration into UAV Relay Systems

- Adapt the hybrid encryption model (AES via MDI-key + OTP via BB84-key) to UAVs links.
- Evaluate the feasibility of the **MPC framework** within the UAV relay scenarios.

Simulation & Performance Analysis

- Develop a simulation environment using Qiskit and Python.
- Incorporate FSO (Free-Space Optical) channel models, accounting for atmospheric loss, turbulence and pointing errors.
- Obtain key performance indicators, primarily the Secret Key Rate (SKR).

Comparative Study of Strategies

- Compare various protocols including MDI-QKD.
- Evaluate hybrid methods like "BB84 + MDI"
- Find the best strategy by analyzing the trade-off between "Communication Capacity" and "Security"

Thank you for listening
